Chapter 0

Introduction to Financial Markets and Fundamental Financial Concepts

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0.1. The different financial markets

0.1.1. Some financial jargons

We first look at some terms that often appear in financial conversations.

- **Buy/sell**: The meanings should be obvious
- **Long/short**: *Long* is just another term for buy, and *short* equals sell
- **Bid:** The price that the buyer is prepared to pay
- **Ask:** The price that the seller is willing to sell

For example, if HSBC stock price is quoted at $98.10/98.15, it means that the bid price is $98.10 and the ask price is $98.15. No trade would occur unless a buyer emerges and is willing to pay the asking price, or a seller emerges who is willing to accept a lower price than the current asking price (i.e. $98.15).

- **Take a "position"**: enter into a trade
- **Close a "position"**: enter into trade(s) to neutralize an existing position

Note that there is a difference between closing an existing position and entering into an offset position with another party. For example, if Bank A starts with a long position in a financial contract with Bank B, and now no longer wants to bear the risk of having this position, it could either: i) go to Bank B and negotiate a close-out price, and close the existing trade; ii) go to Bank C and enter into an opposite trade with the same contract terms. However, in case (ii), while Bank A may not be further exposed to market fluctuations, technically it will still have two trades remaining in its portfolio (with both Bank B and Bank C) and some residual risks remain (e.g. it is possible that one of the parties can go bankrupt and would not honor the contract).

- **Risk (風險):** This is a complex concept, and soon we will look at some of the more refined definitions. Basically we can say that if there is uncertainty in the outcome of an investment, with the possibility of suffering a loss, then the position is a risky position.
- **Hedge (對沖):** Engage in a trading strategy to reduce the fluctuation in profit or loss. Normally this cannot be perfectly done unless identical and opposite positions are entered. Note that in the financial market there is a kind of company called the "Hedge Fund" (對沖基金), which may not hedge most of its positions at all.
- **Arbitrage（套戥）:** Engage in a trading strategy which is supposed to generate a "riskless" profit, i.e. no matter how the market will move in the future, a guaranteed profit can be made. Usually this is achieved by simultaneously entering into multiple trades in order to capture the inefficiencies in the markets. Occasionally such opportunities can arise.
- **"P&L":** A short form of "profit and loss"

0.1.2. Types of financial markets and their characteristics

The four major types of financial markets (or asset classes) are:

- Currency (Foreign Exchange, or FX) market 外匯: e.g. US dollar against Euro
- Fixed Income market (including the interest rate market) 定息工具
    - A significant sub-market under this area is credit trading 信貸
- Equity (or stock) market 股票
- Commodities market 商品 : e.g. gold, silver, coffee

The most active market is the currency market (in terms of monetary amount), followed by the fixed income market. Stock markets are not as active as currency and fixed income, although more people pay attention to the stock market than the others. The markets in the United States are the biggest in the world. In terms of total capitalization (i.e. size of the companies), the combined size of the New York Stock Exchange & NASDAQ was US$ 64.6 trillion (June 2025), whereas a typical daily turnover in NASDAQ is more than US$150 billion (June 2025). In the fixed income market, the amount of all outstanding US government bonds (Treasuries) was US$ 29.0 trillion (July 2025), with daily turnover around US$1.08 trillion (YTD July 2025). For the US corporate bond market: outstanding notional was US$ 11.4 trillion (Q1 2025) and average daily turnover US$59.5 billion (July 2025). The average daily turnover in the FX market globally was US$7.5 trillion (April 2022). It is noted that 88% of all trades involved the US dollar.

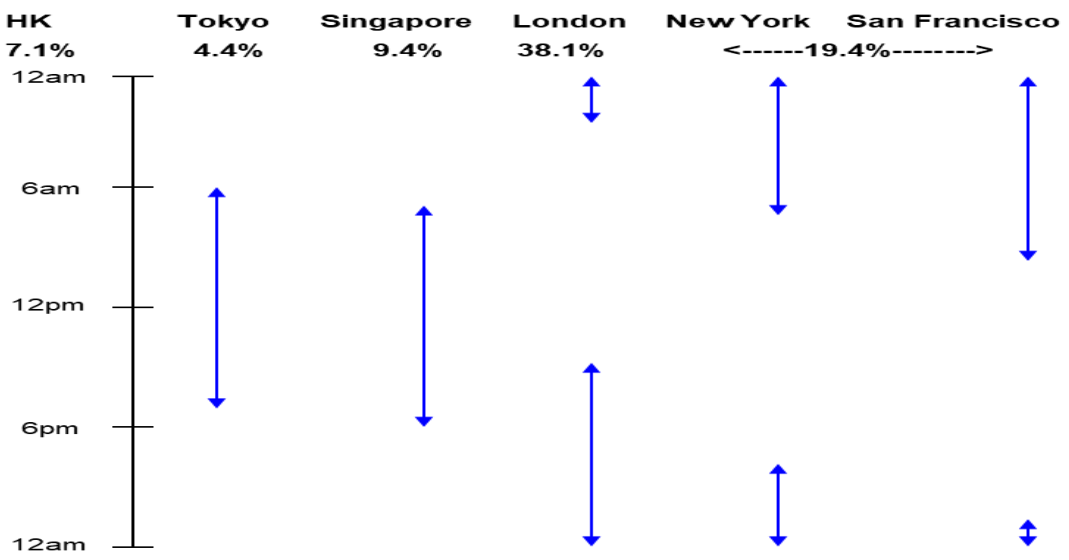
As a comparison, we also give some statistics for the markets in Hong Kong and China. Total market capitalization of the Hong Kong listed stocks was HK$ 44.9 trillion (July 2025, ranked 6th in the world), and the average daily turnover was HK$ 243.7 billion (YTD July 2025). The total market capitalization of Chinese listed stocks in Shenzhen and Shanghai was RMB 94.07 trillion (July 2025, ranked 3rd and 8th in the world), and the average daily turnover was around RMB 1.792 trillion (YTD July 2025). In the Hong Kong interest rate market, the total outstanding amount of Exchange Fund Bills and Notes was HK$ 1.33 trillion (July 2025), with average daily turnover HK$ 11.68 billion (July 2025). If we look at the FX activities in Hong Kong, the daily turnover was US$ 694 billion (April 2022).[1]

Another category which has seen tremendous growth in the last thirty years is the derivatives market 衍生工具市場. However, its nature is different from the markets listed above, because derivatives is a sub-group which exists under each of the asset market, i.e. there can be currency derivatives, interest rate derivatives, equity derivatives, and commodity derivatives, and some derivatives can involve more than one asset class.

0.1.2.1. Currency Market

The global currency market consists of two main sectors: spot and forward markets. The main players are the central banks, major FX trading houses (mostly banks), and some customer demands. The *spot FX* market is a truly global and 24-hour market, because trading activities for the active currency pairs are not limited to one location, which means that the market is always open during the day.

[1] Sources: Bank for International Settlements; SIFMA; Hong Kong Exchange; Hong Kong Monetary Authority



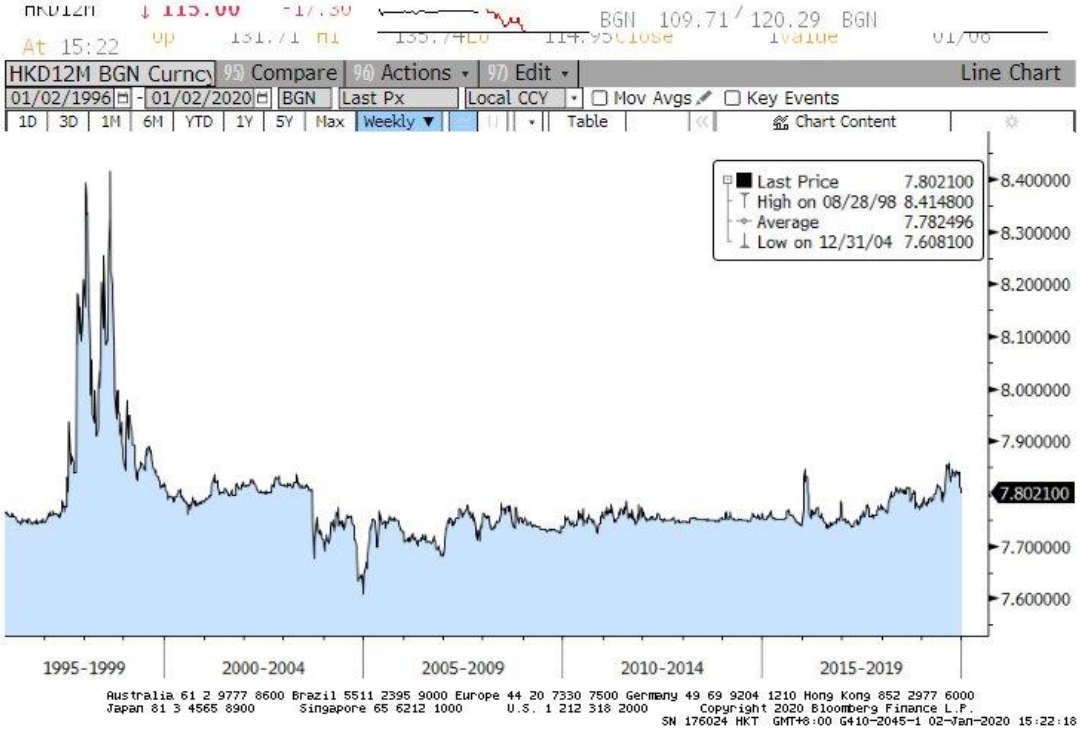
A typical trading day lasts for 11 hours in a single location. It may start in the Asian time zone at 7am in Tokyo (6am in Hong Kong; Singapore is in the same time zone as Hong Kong). In the afternoon in Asia, the European markets will open, and the most significant centre is London (7am there is 3pm in Hong Kong). During the London afternoon, it is the morning in the US market (7am in New York is 8pm in Hong Kong and 3pm in London). The US late afternoon will again overlap with the morning in Asia on the next trading day. The percentages under each centre in the above diagram represent the market share of these centers as of year 2022. London dominates this market, partly because of its advantage in geographical location, having significant overlaps with both Asia and US markets during the normal trading hours. Finally, the market is open on almost every week-day during the year. The only "global" holiday where no trading occurs is January 1.

Spot FX trading is a contract between two parties, i.e. the trading is not conducted through any intermediate party (c.f. an Exchange, as in the case of stocks). In a typical trade, one of the parties will buy currency A and sell currency B at an agreed exchange rate at a certain notional amount. The trade is to be settled at *spot date*, usually two business days after the trade date. For liquid currency pairs, this trade is closest to the theoretical "perfect" market condition, and the bid/ask spread is very tight. For example, we can see a price of US dollar (USD) against Japanese Yen (JPY) at a price quote of 146.75/77. Note that the quoting conventions for different currency pairs are fixed. Say if we see a rate for Australian dollars (AUD) at 0.6550, we understand that the convention is for AUD 1 = USD 0.6550, not the other way round. Of course, we can express this price as USD 1 = AUD 1/0.6550 = AUD 1.5267, but no one in the market will quote a price like that.

Instead of the trade being settled at spot date, it can be settled at a future date, known as the *maturity date* (normally less than one year from today). This type of trade is known as the *forward FX*. A key characteristic is that there is no cost to enter into this trade on the trade date if it is traded at the market quoted price. Because of this feature, it is very convenient to use it as a speculation tool, as in the following example. Hong Kong dollar is pegged to the US dollar at USD 1 to HKD 7.80 since 1983. Supposed that some speculators think that the peg would be changed, and that HK Dollar would appreciate (go up in value). One way to express this view is to enter into a spot FX position, by buying HKD and selling USD. However this strategy requires substantial commitment in USD

upfront. On the other hand, a typical strategy is to enter into a forward transaction, to buy HKD in (say) 6 months. There is no initial amount required, and the position could be closed out at any time before maturity.

USD/HKD 12-month FX forward (Data as of January 2, 2020; source: Bloomberg LP)



From the data above, we can see that while the HK dollar is fixed at the pegged rate, the forward price can still fluctuate a lot. During the Asian financial crisis in 1998, the 12-month HKD forward actually traded at around USD 1 = HKD 8.5000; while in late 2004, it traded at around USD 1 = HKD 7.6200.

USD/CNY FX forward (Data as of January 2, 2020; source: Bloomberg LP)

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USDCNH 99) Chart Spot/Forwards Monitor

China Offshore Spot

Market Type Regular Source BGN Bloomberg BGN

Term	Pts Time	Pts Bid	Pts Ask	Spread	Outrt Bid	Outrt Ask	Time
1) SPOT	15:26	6.9621	6.9626	0.0005	6.9621	6.9626	15:26
2) ON*	15:25	-0.55	1.15	1.70	6.9617	6.9626	15:25
3) TN*	15:26	0.60	2.60	2.00	6.9618	6.9625	15:26
4) SN	15:26	-0.62	1.62	2.24	6.9620	6.9628	15:26
5) 1W	15:24	2.16	8.84	6.68	6.9623	6.9635	15:24
6) 2W	15:26	11.78	17.84	6.06	6.9633	6.9644	15:26
7) 3W	15:11	32.44	35.56	3.12	6.9653	6.9662	15:11
8) 1M	15:26	38.91	51.09	12.18	6.9660	6.9677	15:26
9) 2M	15:26	86.81	101.19	14.38	6.9708	6.9727	15:26
10) 3M	15:26	136.98	153.02	16.04	6.9758	6.9779	15:26
11) 4M							
12) 5M							
13) 6M	15:26	279.92	301.43	21.51	6.9901	6.9927	15:26
14) 9M	15:26	440.39	469.61	29.22	7.0061	7.0096	15:26
15) 12M	15:26	609.32	638.08	28.76	7.0230	7.0264	15:26
16) 15M							
17) 18M							
18) 2Y	15:26	1,400.98	1,467.19	66.21	7.1022	7.1093	15:26

*Near Leg Outrights Precision 4

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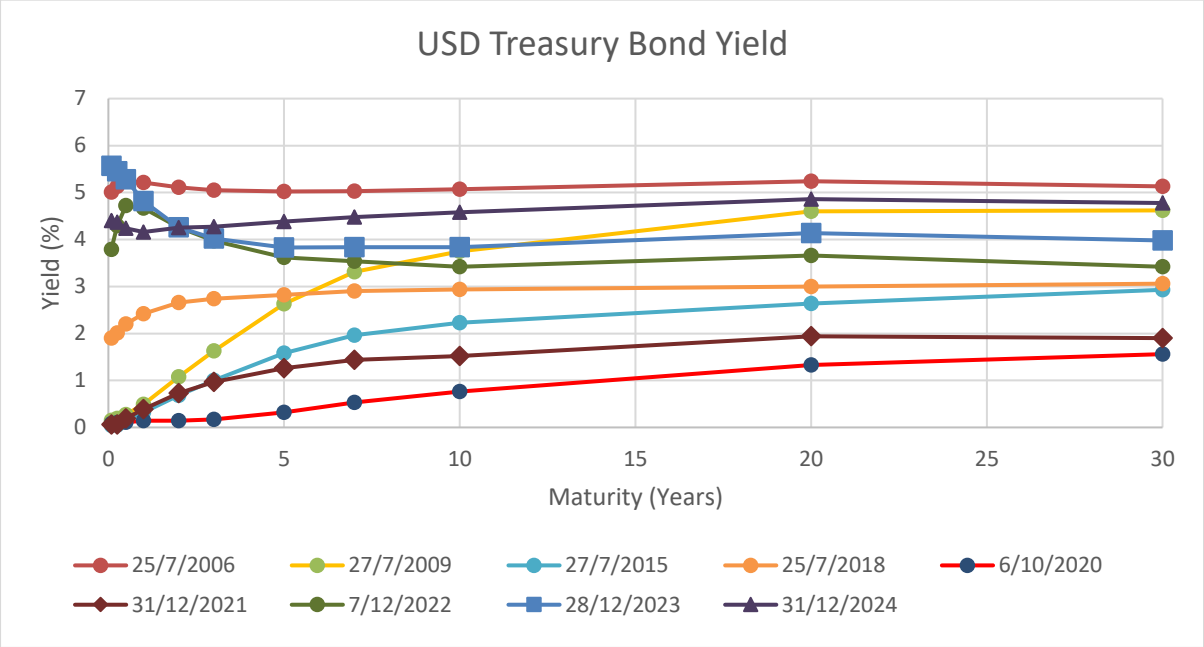
Japan 81 3 4565 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2020 Bloomberg Finance L.P.
SN 176024 HKT GMT+8:00 G410-2045-1 02-Jan-2020 15:26:45

In an earlier example, for the Chinese Yuan, the market consensus in January 2020 was that the currency will weaken in the next year, reversing the trend of expecting the Yuan to appreciate against the USD in the last few years (but note that the actual rate was around 6.70 in Oct 2020).

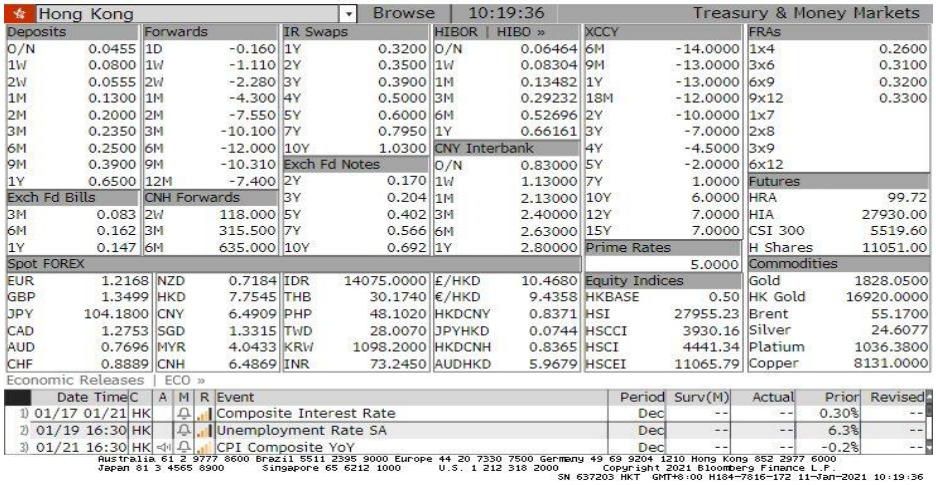
0.1.2.2. Fixed Income market

Interest rate instruments are a means for governments and companies to borrow money. There are many different kinds of interest-rate related instruments, so one must be careful with the meaning of the quoted rate. For example, it is not enough to say that 1-year interest rate of USD is 0.75%, because the rate can refer to a few different types of financial instruments.



In one of the sectors, often known as the fixed income market, the typical instruments are bills, notes and bonds. These terms usually refer to financial instruments issued by governments. The contract terms of these instruments are similar, and the only difference is the time to maturity: a bill has a maturity date of less than 1 year, a note's maturity date is between 1 to 10 years, and the maturity for bonds is 10 years and above, which can sometimes go to 30 years or beyond. Examples of government-issued bonds include the US Treasuries and the Exchange Fund Bills and Notes from Hong Kong. Typical USD Treasury Bond yields from different periods are shown in the diagram above. We can see that the yield curves can take different shapes: upward sloping, mostly flat, or even inverted. (Source: https://home.treasury.gov/resource-center/data-chart-center/interest-rates/TextView?type=daily_treasury_yield_curve&field_tdr_date_value=2024)

A distinction is made regarding the maturity of the instruments. The instruments with maturity less than one year constitute the Money Markets. Typical instruments include the interbank market loans and deposits. Alternatively, instruments with maturity longer than one year constitute the Capital Markets. Government and corporate bonds are most popular. Some HKD market instruments are shown below (data as of January 11, 2021; source: Bloomberg LP):



Hong Kong | Browse | 10:19:36 | Treasury & Money Markets

Deposits		Forwards		IR Swaps		HIBOR | HIBO »		XCCY		FRAs	
O/N	0.0455	1D	-0.160	1Y	0.3200	O/N	0.06464	6M	-14.0000	1x4	0.2600
1W	0.0800	1W	-1.110	2Y	0.3500	1W	0.08304	9M	-13.0000	3x6	0.3100
2W	0.0555	2W	-2.280	3Y	0.3900	1M	0.13482	1Y	-13.0000	6x9	0.3200
1M	0.1300	1M	-4.300	4Y	0.5000	3M	0.29232	18M	-12.0000	9x12	0.3300
2M	0.2000	2M	-7.550	5Y	0.6000	6M	0.52696	2Y	-10.0000	1x7	
3M	0.2350	3M	-10.100	7Y	0.7950	1Y	0.66161	3Y	-7.0000	2x8	
6M	0.2500	6M	-12.000	10Y	1.0300	CNY Interbank		4Y	-4.5000	3x9	
9M	0.3900	9M	-10.310	Exch Fd Notes		O/N	0.83000	5Y	-2.0000	6x12	
1Y	0.6500	12M	-7.400	2Y	0.170	1W	1.13000	7Y	1.0000	Futures	
Exch Fd Bills		CNH Forwards		3Y	0.204	1M	2.13000	10Y	6.0000	HRA	99.72
3M	0.083	2W	118.000	5Y	0.402	3M	2.40000	12Y	7.0000	HIA	27930.00
6M	0.162	3M	315.500	7Y	0.566	6M	2.63000	15Y	7.0000	CSI 300	5519.60
1Y	0.147	6M	635.000	10Y	0.692	1Y	2.80000	Prime Rates		H Shares	11051.00
Spot FOREX									5.0000	Commodities	
EUR	1.2168	NZD	0.7184	IDR	14075.0000	£/HKD	10.4680	Equity Indices		Gold	1828.0500
GBP	1.3499	HKD	7.7545	THB	30.1740	€/HKD	9.4358	HKBASE	0.50	HK Gold	16920.0000
JPY	104.1800	CNY	6.4909	PHP	48.1020	HKDCNY	0.8371	HSI	27955.23	Brent	55.1700
CAD	1.2753	SGD	1.3315	TWD	28.0070	JPYHKD	0.0744	HSCCI	3930.16	Silver	24.6077
AUD	0.7696	MYR	4.0433	KRW	1098.2000	HKDCNH	0.8365	HSCI	4441.34	Platium	1036.3800
CHF	0.8889	CNH	6.4869	INR	73.2450	AUDHKD	5.9679	HSCEI	11065.79	Copper	8131.0000

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	Date Time	C	A	M	R	Event	Period	Surv(M)	Actual	Prior	Revised
1)	01/17 01/21	HK				Composite Interest Rate	Dec	--	--	0.30%	--
2)	01/19 16:30	HK				Unemployment Rate SA	Dec	--	--	6.3%	--
3)	01/21 16:30	HK				CPI Composite YoY	Dec	--	--	-0.2%	--

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Japan 81 3 4565 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2021 Bloomberg Finance L.P.
SN 637203 HKT GMT+8:00 H184-7816-172 11-Jan-2021 10:19:36

Instruments issued by corporates (i.e. private companies) are usually known as *bonds*, irrespective of the time to maturity. An example of a typical bond has the following terms:

- Issuer: Tencent Holdings
- Currency: USD
- Issued Date: June 3, 2020
- Maturity: January 26, 2026
- Coupon: 1.810% per annum
- Payment Frequency: Semi-annual
- Face Value: USD 1000
- Credit Rating: A1 / A+
- Bid price (as of 2/1/2024) = 93.71, equivalent to a yield of around 5.06%

Since the bond was issued a few years ago, the coupon does not reflect the current market conditions. Investors' return should be measured by the yield, which should be calculated by the current price together with the coupon.

0.1.2.3. Equity market

We will mostly use the equity market as an example in the remaining parts of this course. Most stock trading are conducted through an exchange, i.e. you are trading (buying or selling) against an exchange, and the exchange will find the opposite party who is willing to sell / buy the stocks from you. While we normally think about trading stocks, actually many kinds of instruments are traded in an exchange (also known as the listed instruments). The most common types are:

- Stocks 股票
- Exchange Traded Funds (ETFs), especially based on Stock Market Index 指數基金
- Some derivative Instruments 衍生工具 e.g. warrants, index futures 窩輪，期指

On the other hand, some instruments are privately negotiated and traded between two contract parties. Generally known as the over-the-counter (OTC) market, examples of these instruments include:

- Mutual Funds 基金
- Some derivative instruments, e.g. options, swaps 期權，掉期合約
- Structured products 結構性產品 e.g. Equity Linked Note 股票掛鈎高息票據

Many derivative instruments and structured products are customized contracts, and we will look at some examples in Chapter 3.

Equity and Debt

Companies have two means of raising capital, by issuing bonds and new shares. Bond investors hold the company's debt. While they have no control over the management of the company, they have claims over the company's assets in the event of liquidation (ranked in order of seniority according to the type of debt that the investors are holding).

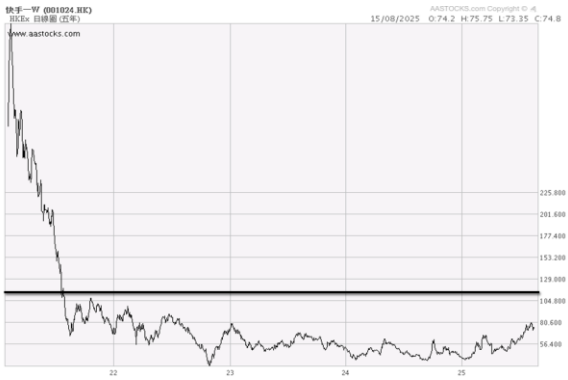
On the other hand, share investors (or shareholders) are the owners of the company. They can participate in the shareholder meetings and would usually have voting rights, which can affect how the company is run (e.g. by electing the board of directors). Under normal circumstances, when the company makes a profit, the shareholders may benefit by means of dividends. However, in the event of liquidation, the shareholders are the last in line of all those who have a claim on the company's assets, namely, after the bond holders and debtors. Unlike partnerships, companies are usually set up to have limited liability, which means that the shareholders are not personally liable to the company's loss. The maximum liability of the shareholders would be the initial amount that they invest to buy the shares.

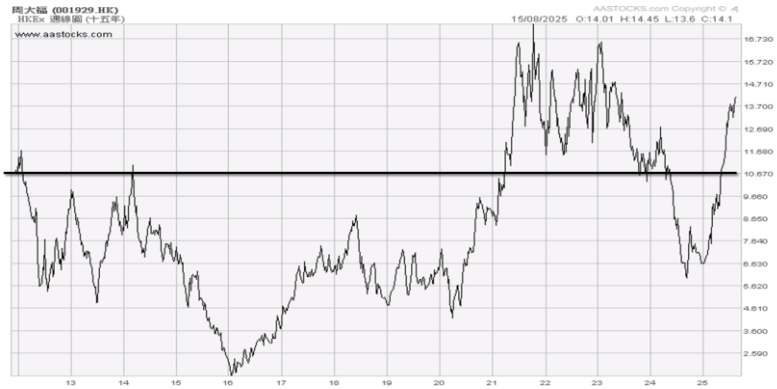
Primary and Secondary Markets

The *primary market* refers to new shares that are traded publicly for the first time. Most of the new shares become available through a process known as the initial public offering (IPO). Sometimes existing shares can also be offered to new investors through private placement. The secondary market is the type of share trading that most people are familiar with. Basically, when investors buy and sell shares in the market, there is a transfer of ownership. As mentioned above, these share transactions are often conducted through a recognized exchange, although it is also possible to arrange a private transfer from one party to another.

Investment banks play a significant role in the initial public offering process. A group of investment banks, collectively known as a syndicate, will often act as the underwriter – they agree to purchase the new shares from the issuing company, and then re-sell them to the public. Whether the IPO will become successful rely on the execution skills of the syndicate. The IPO price is critical: the price needs to be attractive to potential investors. At the same time, the issuing company would want to raise as much capital as possible, and thus they are unwilling to sell the shares at a low price. During the book building period, investment banks collect the responses of interested parties, and the solicitation can be conducted internationally.

To see why IPO pricing can be difficult, we may look at the performance of the biggest IPO companies in Hong Kong in 2021 and 2023 (left – Kuaishou Technology (1024), IPO date 5/2/21, IPO price $115; right – ZJLD Group (6979), IPO date 27/4/23, IPO price $10.82; data up to 16/8/25; source: www.aastocks.com):





Another example shown above is Chow Tai Fook Jewellery Group (1929) (bottom diagram). It was first listed in Hong Kong in December 2011, but the share price had been lower than the IPO price for many years. It was not until mid-2021 when the share price finally rose above the IPO price.

In addition to the local markets, companies may decide to list the shares in a foreign market. As the US market is the most developed and active equity market in the world, many companies would want to be traded there. Applying for IPO in different markets require satisfying different set of listing rules, and the shares are traded by different group of investors. There are more than 200 Chinese companies listed in the US market. These shares are denominated in USD, and are called American Depositary Receipt (ADR). For example, Alibaba was first listed in the US in 2014. Subsequently, it was also listed in Hong Kong in 2019. As US stock market and Hong Kong stock market are opened during different hours (because of time zone difference), investors who have access to both markets can almost trade the share at most times during the day, although there are specific rules concerning the conversion of shares from one market to another.

In recent years, the biggest IPO markets are in China, and Shanghai and Shenzhen have taken the first two spots in the annual international ranking in terms of total size of the IPO. Hong Kong has taken the top spot in the past (as recent as 2019), but it dropped to 3rd or 4th since 2019. Market activities increased more recently and it is forecast to rank again as the biggest IPO market in 2025 (as of August 2025).

Examples of Corporate Actions

Share price reflects the amount that the investors pay or receive when transactions are carried out. One of the fundamental concepts in finance is that share price will only change when new information arrives. However, the nominal share price can also change

because of some technical factors. While the nominal share price will change, it may not cause any profit or loss for the investors. Having said that, one needs to pay attention to these corporate actions, especially when one wants to carry out any numerical analysis based on the share prices. Many systems will adjust the share prices so as to reflect the normal type of pricing actions.

When we refer to the share price, it actually represents the price of one share. The market capitalization of the company is the total number of shares outstanding multiplied by the share price. For various reasons, the company may decide to change the unit size (and the total number of shares outstanding), by splitting into smaller units (**stock split**), or combined to bigger units (**reverse split**). For example:

- Ping An Insurance (stock code 2318) 2-for-1 split on July 27, 2015
    - If you hold 2,000 shares on July 24, 2015 (closing price HKD 97.05), you would hold 4,000 shares from July 27, 2015 onwards. In the morning of July 27, 2015, before the market opened, you would see that the share price of Ping An Insurance would be shown as HKD 48.525 (=97.05/2). The closing price on July 27 was HKD 47.15, i.e. 2.83% drop in price compared to the previous trading day.
- China Tower (stock code 0788) 1-for-10 consolidation on February 20, 2025
    - If you hold 5,000 shares on Feb 19, 2025 (closing price HKD 1.21), you would hold 500 shares from Feb 20, 2025 onwards. The closing price on Feb 20 was HKD 11.88, i.e. 1.82% decrease in price compared to the previous trading day.

Another type of corporate action is a **rights issue**. This is an offer to existing shareholders to buy newly issued shares, usually at a discounted price. Through this action, the company would be able to raise new share capital (similar to the effect of IPO).

As an example, the share price of company ABC is $10. An 1-for-5 rights issue, at price $4 per share, means that current shareholders could buy 1 share for 5 shares that they hold, at a price of $4 per share. If a shareholder has 1000 shares as of the ex-rights date, the entitlement is 200 shares.

To see how this corporate action can affect the theoretical share price, we can calculate the market capitalization of the company. Let's say the total number shares outstanding for this company is 50,000,000, and the share price is $10. If all shareholders take up the 1-for-5 rights issue at $4, the number of new shares issued is 10,000,000, and the new capital is 50,000,000 x $10 + 10,000,000 x $4 = $540,000,000 for 60,000,000 shares, or $9 per share.

A shareholder who starts with 1000 shares (at $10) and subscribes to the offer (i.e. pay an additional 200 x 4 = $800) would end up with 1200 shares at $9 per share. If the share price remains at this level, there is no P&L impact. However, some shareholders may not want to subscribe to the offer, because they don't want to invest more money. The problem is, if most shareholders have subscribed, price adjustment would occur. In the example above, if the shareholder with 1000 shares is the only one not taking up the offer, the share price would drop from $10 to a price very close to $9.[2] It means that the shareholder would suffer a loss of almost 10% after the rights issue.The price drop due to the rights issue is called the dilution effect. To avoid suffering a loss, existing shareholders are "forced" to pay more cash and invest.

[2] Total new capital is 50,000,000 x $10 + 9,999,800 x $4 = $539,999,200 for 9,999,800 share, or $9.0000167 per share.

Stock split and rights issue do not occur very frequently. Another type of corporate action, the payment of **dividends**, is routine. Companies often distribute the profits through dividends to the shareholders. Established companies may pay four dividends per year, and many companies would make two dividend payments per year.

If you are a shareholder on the *record date*, you are entitled to the dividend. For simplicity, let's assume that a company would pay a single cash dividend of $D per share per year, and the current stock price is $P. On the *ex-dividend date*, the share price would drop to $P–D. $D (multiplied by the number of shares held) would be paid to the shareholder on the *dividend payment date*. We could calculate the dividend yield as: D/P x 100%. However, the dividend is only a real return if stock price goes back up to the price before ex-dividend, otherwise there is no P&L impact to the shareholder.

939 HK HKD 6.01 +0.08 H6.01/6.02H 3.3Mx4.5M
At 10:28 Vol 103,083,663 O 5.88H H 6.05H L 5.88H Val 618.184M
939 HK Equity | Settings | Dividend/Split Summary
China Construction Bank Corp
Range 2011 - 2020 | Adjust for Capital Change | Currency As Reported

12 Month Yield	5.84%
Indicated Yield	5.84%
1 Yr Dividend Growth	1.04%
3 Yr Dividend Growth	3.35%
5 Yr Dividend Growth	-1.36%
Last Price	HKD 6.01
Payment Frequency	Annual

Type All | Color Legend | Show Comparative BDVD Forecasts

	Declaration	Ex Date	Record	Payable	Curr	Gross Amt	Type
11)	03/29/21	07/05/21			CNY	.34	BDVD Forecast
12)	03/29/20	07/02/20	07/03/20	07/30/20	CNY	.32	Final
13)	03/27/19	07/02/19	07/03/19	07/30/19	CNY	.306	Final
14)	03/27/18	07/09/18	07/10/18	08/06/18	CNY	.291	Final
15)	03/29/17	06/22/17	06/23/17	07/20/17	CNY	.278	Final
16)	03/30/16	06/22/16	06/23/16	07/22/16	CNY	.274	Final
17)	03/27/15	06/23/15	06/24/15	07/24/15	CNY	.301	Final
18)	03/30/14	07/02/14	07/03/14	08/08/14	CNY	.30	Final
19)	03/24/13	06/13/13	06/14/13	07/12/13	CNY	.268	Final
20)	03/25/12	06/14/12	06/15/12	07/13/12	CNY	.2365	Final
21)	08/21/11	11/07/11			CNY		Omitted
22)	03/27/11	06/16/11	06/17/11	07/15/11	CNY	.2122	Final

3 HK HKD 11.58 -0.02 H11.58/11.60H 32000x208k
At 10:28 Vol 3,754,001 O 11.70H H 11.76H L 11.56H Val 43.727M
3 HK Equity | Settings | Dividend/Split Summary
Hong Kong & China Gas Co Ltd
Range 2011 - 2020 | Adjust for Capital Change | Currency As Reported

12 Month Yield	2.93%
Indicated Yield	2.93%
1 Yr Dividend Growth	8.18%
3 Yr Dividend Growth	9.39%
5 Yr Dividend Growth	9.63%
Last Price	HKD 11.58
Payment Frequency	Semi-Annual

Type All | Color Legend | Show Comparative BDVD Forecasts

	Declaration	Ex Date	Record	Payable	Curr	Gross Amt	Type
11)	03/19/21	06/10/21			HKD	.23	BDVD Forecast
12)	08/17/20	08/31/20	09/01/20	09/14/20	HKD	.12	Interim
13)	03/18/20	06/09/20	06/10/20	06/23/20		5.000%	Bonus
14)	03/19/20	06/09/20	06/10/20	06/23/20	HKD	.219048	Final
15)	08/20/19	09/10/19	09/11/19	10/02/19	HKD	.114286	Interim
16)	03/20/19	05/30/19	05/31/19	06/13/19		10.000%	Bonus
17)	03/20/19	05/30/19	05/31/19	06/13/19	HKD	.199134	Final
18)	08/21/18	09/11/18	09/12/18	10/02/18	HKD	.103896	Interim
19)	03/20/18	06/08/18	06/11/18	06/22/18		10.000%	Bonus
20)	03/20/18	06/08/18	06/11/18	06/22/18	HKD	.181031	Final
21)	08/21/17	09/12/17	09/13/17	10/03/17	HKD	.094451	Interim
22)	03/16/17	06/09/17	06/12/17	06/23/17		10.000%	Bonus
23)	03/16/17	06/09/17	06/12/17	06/23/17	HKD	.164574	Final

Source: Bloomberg LP

The left table above shows the historical dividends of China Construction Bank (0939.HK) from 2011 to 2021. The closing price of the stock on June 30, 2020 was $6.27. The theoretical price before market opened on July 2, 2020 would be $6.27 – 0.35 = $5.92 (assume CNY 0.32 = HKD 0.35). The closing price on July 2, 2020 was $6.09, which should be considered as **going up** (compared to the previous day), rather than coming down.

The right table shows the historical dividends of HK & China Gas (0003.HK). Dividends can be paid as *bonus shares*. In this example, on June 9, 2020, there will be a bonus dividend of 5%. If you hold 2000 shares of HK & China Gas on June 8, 2020, you are entitled to have an additional 100 shares, i.e. a total of 2100 shares on June 9, 2020 (the extra shares would only be received on June 23, 2020, *the payable date*).[3]

The closing price on June 8, 2020 was HKD 13.70. As there is 1 bonus share for 20 shares held, after the ex-dividend date, the share price should drop to 20/21 of the previous day's close (+ share price drop due to cash dividend in this case, which is $0.23 per share). The theoretical price is thus $(13.7–0.23)x20/21= $12.829. As the closing price on June 9, 2020 was HKD 12.86, again the price should be considered as going up compared to the previous day.

[3] The ex-dividend date and record date are different. It is because of a settlement issue. In Hong Kong, when shares are traded, it takes 2 business days for the trade to be settled (i.e.T+2). If you buy shares on June 8, 2020, the shares would only be received on June 10, 2020, when the official record would be updated. Therefore, in order to receive the dividend, one must buy the shares the date before the ex-dividend date, which means that the record date is *after* the ex-dividend date.

A simple way of making money in the stock market

Finally, we look at a simple example regarding the different ways of making money in the stock market. Let's say HSBC shares are trading at HKD 64 per share when you enter into a position.

- If you are **bullish** (i.e. you think the shares would go up), then you should go long
- If you are **bearish** (i.e. you think the shares would go down), you should go short

Consider the following positions and scenarios:

Position (shares)	P&L	
	Share price = $60	Share price = $70
Long 10000	–$40000	+$60000
Short 4000	+$16000	–$24000

We can easily see that we can make money if we:

- Take a long position in stocks, and share price goes up;
- Take a short position in stocks, and share price goes down (note there may be regulations regarding short sell and not all shares in the markets can be short).

Alternatively, a more sophisticated trade is to construct a "synthetic" position in stocks, and hopefully the market view is correct. For example, by taking a long position in a stock index (say Hang Seng Index or the Tracker Fund (2800)) and short most of the component shares in the index, it can be set up as equivalent to holding a long position in some of the constituent stocks. This is an example of a "fixed" relationship between various instruments. Some such relationships exist, and trading opportunities may arise if the relationships are not obeyed (refer to the Index Arbitrage strategy in Chapter 6).

0.2. Fundamental financial concepts

0.2.1. Risk and return

Formulation of financial theory depends critically on the concept of the relationship between risk and return. In the example below, which strategy would you choose?

Strategy A:

- 50% chance of earning a 10% return
- 50% chance of losing 5%

Strategy B:

- 50% chance of earning a 15% return
- 50% chance of losing 12%

We can easily calculate the expected return of *A* is 2.5% (0.5 x 0.1 + 0.5 x (–0.05)) and for *B* is 1.5%. However, one may argue that the (high) possibility of losing 12% seems to be a bit excessive, and so strategy *B* may not be attractive. From this simple example, we can see that a higher absolute return may not always be the best strategy; we need to consider the expected return AND risk (the possibility of loss) together.

Many important finance theories are based on the fact that people make rational decisions. A rational investor always wants to achieve *the highest return with the minimum risk.* But there have been challenges to this simple and beautiful theory. Let's look at the following scenarios.

Situation 1:
Expected return of a lottery is $10; there is a small chance that you would earn nothing. It costs you $9 to play this game. How many times would you want to play?

Situation 2:
Expected return of an investment is $10 Million; there is a small chance that you would earn nothing, and it costs you $9 Million to invest in this project. How many times would you want to invest?

The expected return in both these examples is about 11% higher than the cost, which means that in the long run, one should expect to make a return of 11%. The question is, would your answer be different in these two situations? Is it because of the high initial cost in the second case, and the fear of a loss of $9 Million makes you reluctant to invest in the choice more frequently?

This has actually been a long standing problem, which was first noticed by Daniel Bernoulli in 1738. In the famous **St. Petersburg Paradox**, a coin is tossed until the first head appears. Let n be the number of times a tail appears before the first head is tossed. In this game, the payout is 2^n. For example, if the sequence goes like this: TTTH, i.e. there are 3 tails before a head appears, the payout is 2^3 = $8. Now how much money would you want to pay to play the game?

To arrive at a mathematically correct answer, we need to calculate the expected payoff, which is given as:

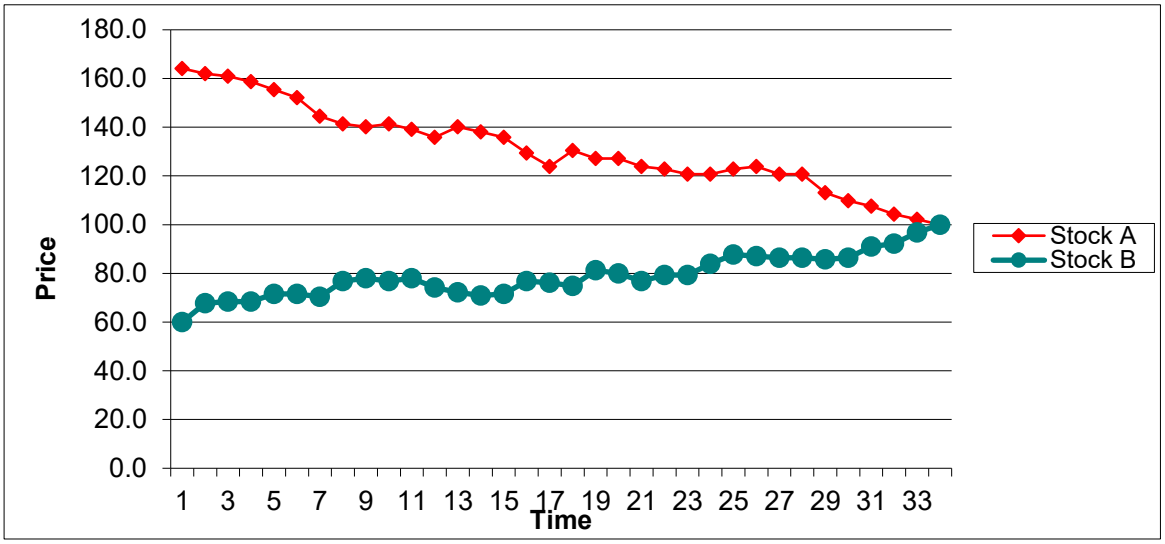
No. of tails before a head appears	Probability	Payoff	Expected payoff
0	$(½)^1$	$\$2^0 = \$1$	½ x $1 = $½
1	$(½)^2$	$\$2^1 = \$2$	$(½)^2$ x $2 = $½
2	$(½)^3$	$\$2^2 = \$4$	$(½)^3$ x $4 = $½
3	$(½)^4$	$\$2^3 = \$8$	$(½)^4$ x $8 = $½
N	$(½)^{(n+1)}$	$\$2^n$	$(½)^{(n+1)}$ x $\$2^n$ = $½

$$\text{Expected payoff} = \sum probability \times payoff = \tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{2} + \cdots = \infty \ !$$

If the game is being played many times, the fair price of this game should be close to the expected payoff, but most people would not pay a lot of money to play because one may be unlucky and cannot realize a big enough gain within the first few times the game is played. The lesson is that even though we may have a correct mathematical model, the actual trading decision is more complex. We will come back and look at a solution to this problem when we talk about utility functions in Chapter 4.

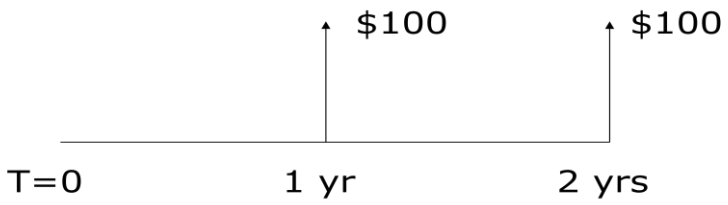
Another problem refers to a commonly encountered situation. Let's say that the stock prices of companies *A* and *B* are both equal to $100 today, i.e. price of stock *A* = price of stock *B* = $100. Furthermore, it is given that the expected standard deviation of price movement of stock *A* is equal to the expected standard deviation of price movement of stock *B* (a technical condition to be explained in the next chapter). Is it more likely for stock *A* or stock *B* to move to $102 tomorrow?

A celebrated finance theory called the Efficient Market Hypothesis (EMH) claims that all currently known information is captured by the stock price, and price movements follow a "random walk" unless new information arrives (c.f. Appendix 1 of these notes). Therefore, the chance of either of these stocks moving to $102 is going to be the same. However, would your answer be different if you are shown the following graph?



0.2.2. Time value of money

How do we compare cashflows occurring at different times?



The horizontal line is a "time line" and the cashflows that are received are represented by arrows lying above the line. In this example, assume that the interest rate is 5% per annum. If you need $*P* to get $100 in 1 year, the formula to calculate $*P* is:

$$100 = P \times (1+5\%)^1, \text{ or } P = \frac{100}{(1+5\%)} = 95.24$$

Similarly, you need $$Q$ to get $100 in 2 years:

$$100 = Q \times (1+5\%)^2, \text{ or } Q = \frac{100}{(1+5\%)^2} = 90.70$$

In general,

$$PV = cashflow \times df \text{ where } df = \frac{1}{(1+r_t)^t}$$

By definition, df is known as the discounting factor at time t, and P above is the present value (PV) of the 1-year cashflow.

0.2.3. Calculation of returns

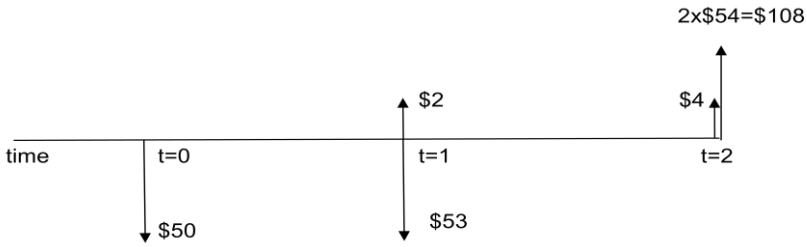
The return of an investment is given by the simple formula: (end value – initial value) / (initial value). For example, if the initial investment is $1000 and after one year, this position is sold at $1200, the return r is:

r = $1200 / $1000 – 1
= 20% per annum, or written as 20% p.a.

Actual calculation of returns can be more complicated. In the case when there are multiple trades occurring at different times, there are different ways to represent the return of the investment.

We start with an example. The portfolio has the following cashflows:

- Bought 1 share at $50 at time 0
- Bought 1 share at $53 at time 1-year
- Sold 2 shares at $54 at time 2-year
- Received $2 dividend at time 1-year
- Received $4 dividend at time 2-year



We can calculate the **dollar-weighted return**, or more commonly known as the **Internal Rate of Return** (IRR) of an investment. This is a representation of the economic reality in terms of the cashflows. By equating the present values at time t=0,

$$50 + \frac{53}{1+r} = \frac{2}{1+r} + \frac{(108+4)}{(1+r)^2} \Rightarrow r = 7.117\%$$

Alternatively, we can ignore the number of shares held in each period and calculate the **time-weighted return**. In the previous example, the return in year 1 = (53+2)/50 – 1 = 10%, and the return in year 2 = (54+2)/53 = 5.66%. The arithmetic average of the 2-year return = (10%+5.66%)/2 = 7.83%, and the geometric average return = $[1.10 \times 1.0566]^{0.5} - 1$ = 7.81%.[4] In an extreme example, say the stock price starts at $100, goes up 100% in year 1 to $200, and drops 50% to $100 in year 2. The arithmetic average return is (100–50)/2 = +25%, while the geometric average return is 0 ($100 becomes $100 in 2 years).

Although it seems that the IRR is a better way to represent the actual investment result of an investor, it may not always be the benchmark quoted in the financial market.

[4] Note that the geometric average must be smaller than the arithmetic average by a well-known mathematical property.

The reason is that while the IRR takes into account of the timing of the cash flows, a fund manager usually does not have any control on this timing (which depends on investors' subscriptions and redemptions). Therefore time-weighted return can be more relevant to the fund manager's skills.

Even if we agree with this logic, we still need to choose between the arithmetic and geometric return. Geometric return takes into account of the holding period, and it calculates the historical performance accurately, i.e. this return accurately represents what the investor would get at the end of the investment period. However, for the prediction of future 1-period performance, the arithmetic average is the unbiased estimator. In the extreme example given earlier, if there is a 50% chance for each scenario, the expected stock price is (200+50)/2 = 125, which represents a return of 25%, as indicated by the arithmetic average return.

To conclude, it is noted that different methods of calculating the investment return exist, and which method is best depends on how the return is to be used in practice.

0.2.4. Compounding conventions

In the simplest calculation of return, the time period is not taken into account. For example, if the initial investment is $1000 and after six months, this position is sold at $1100, we can say that the return of this investment is 1100/1000 – 1 = 10%. However, if we have another investment and it achieves a return of 10% in 2 months, normally we would say that the second investment is superior. For a valid comparison, it is more appropriate to express the ***annualized return*** of the investment, i.e. the return is given as if it is accumulated for one year. If r_s is the annualized return of an investment, the correct formula is then:

(end value – initial value) / (initial value) = $r_s\, t$

where t is the time (expressed in years). In the second example above, 10% = r_sx(2/12), or r_s =60% p.a.

The above expression is not the only possible calculation. As seen in the example in section 0.1.2.2, there are many kinds of interest rates in the financial markets. For accurate calculations, we must use the correct expression in calculating the cash flows according to the ***compounding convention of the interest rate***. It refers to the different ways in which the time t is used.

Let r_s, r_a, r_{sa}, r_c be interest rates. There are four common compounding conventions:

Simple interest:	$(1+ r_s\, t)$
Annual compounding:	$(1+ r_a)^{\wedge}\, t$
Semi-annual compounding:	$(1+ r_{sa}/2)^{\wedge}(2t)$
Continuous compounding:	$(\exp(r_c\, t))$

For example, say we have $1000 today, and the interest rates are given as follows:

(1) 6-month rate is 4.8%p.a., simple interest convention
- The cash amount after 6 months is: $1000 x (1+0.048 x 0.5) = $1024

(2) 2-year interest rate is 5%p.a., annual compounding convention
- The cash amount after 2 years is: $1000 x $(1+0.05)^2$ = $1102.5

(3) 2-year interest rate is 5.1%p.a., semi-annual compounding convention
- The cash amount after 2 years is: $1000 x $(1+0.051/2)^{2x2}$ = $1105.96

(4) 3-year interest rate is 5.3%p.a., continuous compounding convention

- The cash amount after 3 years is: $1000 x $e^{0.053 \times 3}$ = $1172.34

We may also use these conventions in expressing the return of an investment. For example, an initial investment of $1000 becomes $1120 after 3 months (i.e. t = 3/12 = 0.25 year). The corresponding returns are:

(1) 1120/1000 = 1+ $r_s\, t$, r_s = 48.0% p.a.
(2) 1120/1000 = (1+ r_a)^ t, r_a = 57.5% p.a.
(3) 1120/1000 = (1+ r_{sa} /2)^(2t), r_{sa} = 50.9% p.a.
(4) 1120/1000 = exp($r_c\, t$), r_c = 45.3% p.a.

Given the above formulas, it is straightforward to convert an interest rate from one convention to another. For example, to convert an interest rate from an annual compounding convention to the continuous compounding convention, we can write:

$$(1+ r_a)^{\wedge}\, t = (\exp(r_c\, t))$$

However, this is only a theoretical calculation. The conventions for market interest rates are fixed and pre-defined. For example, deposit rates (less than one year maturity) are always calculated in the simple interest convention. Note that while the continuously compounded rate is often quoted in textbooks, it is never used in financial markets.

In the financial markets, there are other kinds of rates, such as swap rates, which require an understanding of the structure of the underlying product (i.e. the swap). This is outside the scope of our current discussion.